LinkedIn: **Mikhail Jacques**
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Cell: 052-507-7979

PROFESSIONAL SUMMARY

- Software engineer with 10+ years at Israel Aerospace Industries, developing real-time, embedded, and desktop software for UAVs, robotics, and laboratory systems.
- Specializing in C/C++ across Windows, Linux, and RTOS environments, with experience from requirements and architecture through testing, integration, and documentation.
- Skilled in leveraging advanced AI tools, conducting in-depth research, writing efficient and high-quality code, and producing detailed, comprehensive technical documentation.

PROFESSIONAL EXPERIENCE

2015 - Present Software Engineer @ Israel Aerospace Industries - Malat, Israel
Developed real-time, embedded, and desktop software for UAVs, robotics, and laboratory systems using C, C++, C#, and Python. Currently leading development of an AI application for aviation document translation and verification.

2015 - 2015 Software Engineer @ Hewlett-Packard, Yehud, Israel
Developed LeanFT, an automated software testing tool, using C# and Java.

2014 - 2014 Software Engineer @ Galil Soft, Bnei Brak, Israel
Developed software for client projects using C++ in a Linux environment.

2012 - 2013 Software Engineer @ TrioSoft, Ashdod, Israel
Developed an embedded multi-mainline irrigation controller for Netafim using C++.

2010 - 2010 Computer Science Lecturer @ Trent University, Ontario, Canada
Developed and delivered information systems courses.

2009 - 2009 Police Officer @ Ontario Provincial Police, Ontario, Canada
Maintained public order, enforced federal and provincial laws, prevented crime, conducted investigations, assisted victims, responded to emergencies, and made arrests.

2007 - 2008 Computer Science Lecturer @ Trent University, Ontario, Canada
Designed and delivered courses in information systems, programming languages, and computer organization.

2002 - 2006 Research and Teaching Assistant @ Trent University, Ontario, Canada
Delivered seminars on behalf of faculty members, developed simulation models for parallel job-scheduling strategies, and co-authored scientific publications.

1997 - 2000 Project Coordinator @ Intel, Kiryat Gat, Israel, New Mexico, Oregon, USA
Coordinated the design, installation, and qualification of electronic silicon-chip manufacturing equipment.

1996 - 1997 Marine Electrical Officer @ Israeli Merchant Fleet, Israel
Maintained safe and reliable operation of shipboard electrical systems.

TECHNICAL SKILLS

- AI tools: Hermes, OpenClaw, GitHub Copilot, OpenAI Codex/ChatGPT, Anthropic Claude Code, Google Gemini/NotebookLM, Grammarly
- Programming Languages: C, C++, C#, Python
- OS: Windows, Ubuntu, embedded Linux, GHS INTEGRITY RTOS
- Design Patterns: GoF design patterns in C++ and C#
- Networking: Boost.Asio, TCP/UDP, RS-232, RTI DDS, SOCKS, CAN bus
- Version Control: Git, MS TFS, SVN
- IDEs: MS Visual Studio, Eclipse, PyCharm, GHS Multi
- Source Code Editors: Visual Studio Code, Qt Creator, Obsidian
- Static Code Analysis Tools: KLOCWORK, PVS-Studio
- Management tools: Microsoft Azure DevOps, Jira
- Auxiliary tools: CMake, MS Office, Beyond Compare, Wireshark, Packet Sender

FORMAL EDUCATION

2008 - 2008 Diploma of Basic Constable Training @ Ontario Police College and Ontario Provincial Police Academy, Ontario, Canada

2004 - 2006 **Master of Science in Computer Science** @ Trent University, Ontario, Canada
Program: Applications of Modeling in the Natural and Social Sciences
Thesis: Workload Modeling and Internal Backfilling for Parallel Job Scheduling

2001 - 2004 Bachelor of Science in Computer Science @ Trent University, Ontario, Canada

1992 - 1993 Diploma of Technician, Marine Electricity and Control @ Mevoot Yam Marine Technological College, Mikhmoret, Israel

1990 - 1992 Matriculation Certificate, Marine Electricity and Control @ ORT Yami, Ashdod, Israel

AWARDS

- Recipient of the NSERC (Natural Sciences and Engineering Research Council of Canada) Postgraduate Scholarship (PGS)

LANGUAGES

- English, Hebrew, Russian

SECURITY

- Top secret (Level 3) security clearance

MILITARY SERVICE

1993 - 1996 Marine Electrician @ Israeli Navy (Shayetet 13)
Maintained safe, reliable operation of naval electrical systems.

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SUMMARY OF PROFESSIONAL ACTIVITY AT ISRAEL AEROSPACE INDUSTRIES (IAI)

Since joining IAI in December 2015, I have consistently developed and implemented advanced software solutions for real-time systems, research, and desktop applications, leading full development cycles from requirements definition through architectural design, implementation, testing, integration, and comprehensive technical documentation.

Airborne Software Department (December 2015 - October 2020)

During this period, I developed new control logic in C, from specification through implementation, working closely with avionics engineers on requirements definition and technical documentation. The logic primarily addressed engine, fuel, and fuel-pump management.
On my own initiative, I undertook an in-depth review and upgraded substantial sections of the UAV control software, which was difficult to maintain. My key activities included:

- Rebuilding and standardizing critical modules: digital inputs and outputs, serial links, nonvolatile memory, inter-computer communication (AVCs), and antenna management, among others.
- Improving performance on PowerPC processors and upgrading the development environment to newer versions of GHS Multi.
- Reorganizing the directory tree and arranging files by topic and device.
- Removing old, inactive, and redundant code; aligning coding styles; standardizing file and variable names; and improving readability and consistency while reducing coupling.
- Developing and integrating new capabilities to improve the reliability and safety of system operation.
- Developing a new avionics monitoring application (Avim) using C++ and Qt.

Algorithms Department (November 2020 - December 2021)

Developed a central management module in C++ on Windows for the Image Landing project, including requirements gathering, design, implementation, testing, and documentation.

Robotics Department - ELTA (January 2022 - June 2022)

Developed a central management module in C++ on Windows for the Worldview project, including managing internal communications among robotic subsystems.

Product Software Department (July 2022 - July 2023)

Conducted technology research and developed experimental applications in C, C++, C#, and Python.

Product Software Department (July 2023 - January 2024)

Developed a real-time production application in C++ for Windows for the GVDM project. The Israeli Air Force currently uses this software.

Product Software Department (February 2024 - December 2025)

Developed a signal-processing application in C++ for a UEI embedded system running Linux.

AI Software Department (January 2026 - Present)

Leading the development of an AI-based application that translates complex aviation technical documentation from Hebrew to English (assembly instructions and inspection cards), precisely preserving the structure and layout of the original PDFs. Designing a multi-stage quality-control framework in which independent verification agents automatically check translations using structural, geometric, OCR, and semantic assessments, with every finding recorded as evidence. As a result, no unverified output reaches human approval. Developing the product into a user-friendly Windows tool that enables subject-matter experts who are not engineers to review findings, enter corrections, and build an increasingly refined glossary across hundreds of documents.

Mentoring and Collaboration

Throughout my years at IAI, I have also contributed by training and mentoring junior software engineers, collaborating with avionics engineers across technical disciplines, and helping others understand complex code and documentation.

Letters of Recommendation and Evaluations

Over the years, I have received numerous letters of recommendation and evaluation reports from clients, colleagues, and managers at IAI, reflecting my commitment, professionalism, and the broad impact of my work.

Kind regards,
Mikhail Jacques
Software Engineer, IAI
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